for Semantic Segmentation

1. What Is This Paper About?

Before FCN, deep learning networks could classify whole images (e.g. 'this is a cat') but could **not label every individual pixel**. FCN solved this by turning a classification network into one that outputs a full segmentation map — assigning a class label to every pixel in one forward pass.

■ Core Goal: Given an image, predict the class of **every pixel** — dog, cat, sky, road, background, etc. This is called **Semantic Segmentation**.

2. Prerequisite Concepts You Must Know

Concept	Plain English Explanation
Convolutional Layer (Conv)	Slides a small filter (e.g. 3×3) over the image to detect local patterns like edges and textures.
Pooling (Max Pool)	Shrinks the feature map by taking the max in a region. Reduces spatial size, increases receptive field.
ReLU Activation	Non-linearity: keeps positive values, zeros out negatives. Applied after each conv layer.
Receptive Field	The region of the original image a neuron 'sees'. Deeper layers see larger regions.
Stride	How many pixels the filter moves per step. Stride 2 = output half the spatial size.
Feature Map	The output tensor after a conv layer. Has H × W × C dimensions (height, width, channels).
Fully Connected (FC) Layer	A dense layer where every input connects to every output. Loses spatial info.
Upsampling / Deconvolution	The reverse of pooling — makes a small feature map larger (e.g. 14×14 → 224×224).
Transfer Learning / Fine-tuning	Start from weights trained on ImageNet (classification), then adapt to segmentation.
Mean IU (Intersection over Union)	Key metric: overlap of predicted mask vs ground truth, averaged over all classes.

3. The Core Idea — Step by Step

Step 1 — The Problem with Classification Networks

Networks like AlexNet, VGG, GoogLeNet end with **fully connected (FC) layers**. FC layers produce a single vector (e.g. 1000 class scores). They throw away all spatial information — you lose where things are in the image.

Step 2 — Convolutionalization (The Key Trick)

■ Key Insight: A fully connected layer over an N×N feature map is **mathematically identical** to a convolution with an N×N kernel. So you can re-interpret any classification network as a fully convolutional one — and it now works on **any image size**!

FC layer (4096 units) → 1×1×4096 conv. This produces a **spatial map** instead of a single score. Now the output tells you where in the image each class is detected.

Step 3 — Upsampling Back to Full Resolution

After convolutionalization, the output is very small (e.g. 7×7 or 14×14). We need to upsample it back to the original image size (e.g. 224×224). FCN uses **deconvolution (transposed convolution)** — a learned upsampling layer that can be trained end-to-end.

Step 4 — Skip Connections (The Secret Weapon)

The deep layers know *what* (semantics) but the shallow layers know *where* (fine detail). FCN combines both with skip connections:

Model	How It Works	Mean IU
FCN-32s	Upsample final layer by 32× in one shot. Blurry.	59.4
FCN-16s	Combine pool4 + final layer, upsample by 16×. Sharper.	62.4
FCN-8s	Add pool3 too, upsample by 8×. Best detail.	62.7 ✓

4. Architecture at a Glance

```
Input Image → [Conv layers: pool1 → pool2 → pool3 → pool4 → pool5] → [fc6
→ fc7 as 1×1 convs] → Coarse prediction (stride 32)
↑ Skip from pool4 → fuse → Finer prediction (stride 16)
↑ Skip from pool3 → fuse → Finest prediction (stride 8) → Output
segmentation map
```

The backbone is VGG-16 (best performer). The FC layers become 1×1 convolutions. The final output has the **same spatial dimensions as the input image**.

5. Training Details

- Optimizer: SGD with momentum 0.9
- Loss: Per-pixel multinomial logistic loss (cross-entropy at every pixel)
- Fine-tuning: Start from ImageNet-pretrained weights; full network fine-tuned end-to-end
- Batch size: 20 images; learning rates 10■³ (AlexNet), 10■■ (VGG), 5×10■■ (GoogLeNet)
- Patchwise sampling: Tried it — whole-image training equally effective and faster
- Class balancing: Not needed (background ~75% — mild imbalance ignored)

- Training time: ~3 days for FCN-32s, ~1 more day each for FCN-16s and FCN-8s

6. Results

FCN-8s achieved **62.2% mean IU** on PASCAL VOC 2012 test set — a **20% relative improvement** over the previous state-of-the-art (SDS at 51.6%). Inference takes ~175 ms vs ~50 s for SDS — a **286× speedup**.

Method	Mean IU (VOC2012)	Inference Time
R-CNN	47.9%	–
SDS (prior SOTA)	51.6%	~50 s
FCN-8s (this paper)	62.2% (+20%)	~175 ms

Also achieved SOTA on NYUDv2 (RGB-D indoor scenes) and SIFT Flow (scene parsing). For NYUDv2, using RGB + HHA depth encoding boosted mean IU from 29.2 to 34.0.

7. Key Innovations (Exam / Interview Points)

■ Fully Convolutional

First work to train an FCN end-to-end for pixelwise prediction from supervised pre-training.

■ Any Input Size

Unlike classification nets, FCN processes images of arbitrary size in one pass.

■ Deconvolution Upsampling

Learnable upsampling (transposed convolution) instead of fixed bilinear interpolation.

■ Skip Architecture

Fusing predictions from shallow and deep layers balances semantics vs detail.

■ Efficiency

Whole-image inference is far faster than patch-by-patch sliding window approaches.

8. Limitations & What Comes Next

✗ Output is still somewhat coarse even with FCN-8s (stride-8 resolution)

✗ No instance separation — all cats are one class, not individual cat-1, cat-2

✗ Does not handle fine boundaries well (inspired U-Net, DeepLab, etc.)

✗ Training from scratch infeasible — requires ImageNet pretraining

✗ Patchwise methods can handle class imbalance better in some cases

9. Simple Mental Model to Remember

Think of it like this:

1. Take a VGG net trained on ImageNet — it's good at 'what is in this image'
2. Remove the final classification head; replace with 1×1 convolutions
3. It now outputs a tiny heatmap (e.g. 7×7) — too small, but knows *what*
4. Use skip connections from shallower layers to recover spatial detail (*where*)
5. Upsample back to full image size using learned deconvolution
6. Result: every pixel has a class label

10. Quick Glossary for Segmentation Projects

Term	Meaning
Semantic Segmentation	Label every pixel with a class (cat, dog, background). No instance separation.
Instance Segmentation	Label each object separately (cat-1, cat-2). Needs Mask R-CNN etc.
Panoptic Segmentation	Combines semantic + instance. Each pixel has a class AND an instance ID.
IoU / Jaccard Index	Area of overlap / Area of union. 1.0 = perfect match. Key evaluation metric.
Mean IU	IoU averaged over all classes including background.
Backbone	The feature extractor (e.g. VGG, ResNet). Usually pretrained on ImageNet.
Encoder	The contracting/downsampling part. Extracts features, reduces resolution.
Decoder	The expanding/upsampling part. Reconstructs spatial resolution for per-pixel prediction.
Skip Connection	A shortcut that passes features from an early layer directly to a later layer.
Deconvolution	Transposed convolution — a learnable way to upsample feature maps.
Bilinear Upsampling	Simple non-learned upsampling using bilinear interpolation.
Fine-tuning	Continue training a pretrained model on a new dataset/task with a small learning rate.

■ **For Your Segmentation Project:** FCN is the foundation. All modern segmentation architectures (U-Net, DeepLab, SegNet, Mask R-CNN) build on FCN's ideas: fully convolutional inference, upsampling, skip connections. Understand FCN deeply and the rest follows naturally.